ELIAS AL ALAM

Environmental Engineer

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Swiss Driving License B | June 30, 1997



EDUCATION

Master of Environmental Science

University of Lausanne, Lausanne, Switzerland

08/2022 - 07/2025 | Switzerland

- Thesis: Investigated the impact of tributary inputs on sediment size distribution downstream of the Grande Dixence dam using high-resolution imagery and photogrammetry techniques.
- Developed a supervised learning model achieving 97% accuracy for detecting oil spills from satellite images.

Bachelor of Mechanical Engineering

American University of Beirut, Byblos, Lebanon

08/2016 - 05/2021 | Lebanon

- Designed and built a cost-effective tribometer to test wear and friction of RZeppa joints, reducing manufacturing costs to 1/40 of commercial alternatives.
- Developed a smart home and automated irrigation system using LabVIEW, enhancing energy efficiency and reducing water consumption by 30%.

LANGUAGES

English
C2 (Proficient)

French
B2 (Upper Intermediate)

Arabic
C2 (Proficient)

German
A1 (Beginner)

PROFESSIONAL EXPERIENCE

Research Assistant (40%)

University of Lausanne, Lausanne, Switzerland

03/2023 - 09/2024

- Analyzed 12 years of glacial elevation data from 6 glaciers, identifying critical erosion points.
- Collaborated with researchers from EPFL, ETH Zurich, and Eawag on a multidisciplinary glacier study.
- Utilized QGIS, ArcGIS, and Python/XDEM to track changes in elevation and proglacial area.

Intern: Geophysical Field Engineer

Geo2x & GEOOL, Yverdon-les-Bains, Switzerland

08/2023 - 09/2023

- Deployed and monitored 5,000 seismic sensors across Switzerland for a national geothermal reservoir mapping project.
- Managed sensor installation and data collection under challenging conditions.

Intern: Mechanical Engineering Technician

DEMCO Steel, Beirut, Lebanon

09/2019 - 06/2019

- Performed inspections and diagnostics on industrial machinery, increasing production efficiency by 25% and reducing downtime by 150 hours per month.

SKILLS

Geomatics & Modeling

QGIS, ArcGIS, Remote Sensing, Environmental Impact Assessment, 3D Modeling, Life Cycle Assessment, Sustainability Analysis, Sustainable Manufacturing, Python/XDEM for Geospatial Analysis, LiDAR Data Processing

Data Analysis & Processing

Statistical Analysis & Visualization, LiDAR Data Processing, Machine Learning & AI, Data Acquisition & Management, Environmental Analysis & Sustainability

Programming & Development

Python, R, Julia, Rust, HTML, CSS, JavaScript, MATLAB

Engineering & Simulation

Thermodynamics & Mechanics, Statics & Dynamics, Finite Element Analysis, Computational Fluid Dynamics, MATLAB, SolidWorks, AutoCAD, Ansys, Fluid Mechanics, Materials Science

SOCIAL MEDIA

Portfolio
https://www.eliasalalam.dev

LinkedIn
https://www.linkedin.com/in/elias-al-alam/

GitHub
https://github.com/michmazbout

Instagram
https://www.instagram.com/elias_el_alam/